

Parallel Image Stitching

Performance Analysis of Concurrent Architectures in Python

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July 20, 2026



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Da un secolo, oltre.

Introduction



- **Image Stitching:** combining multiple overlapping images into a single high-resolution panorama
- **Applications:** geospatial mapping, autonomous driving, virtual reality, consumer photography
- **The Problem:** the standard sequential pipeline scales poorly as resolution and dataset size grow
- **Objective:** evaluate Python concurrency paradigms against two language-specific bottlenecks:
 - The **Global Interpreter Lock (GIL)** restricts true multi-threaded execution
 - **Inter-Process Communication (IPC)** introduces serialization overhead

Visual Overview

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- **Dataset:** OpenDroneMap aerial imagery, downsampled to 50% resolution
- **Sliding window:** panoramas built from non-overlapping windows of 4 images at a time
- **Six architectures** benchmarked on this exact pipeline against a sequential baseline

The Stitching Pipeline: Feature Processing

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1. Feature Extraction (SIFT):

- Detects scale-space extrema via the Difference of Gaussians
- Runs independently per image, so it is embarrassingly parallel

2. Feature Matching (FLANN):

- Compares 128-dimensional descriptors between adjacent frames
- Lowe's ratio test with $\tau < 0.7$ filters out spurious matches

3. Homography Estimation (RANSAC):

- Iteratively fits a 3×3 matrix H that separates inliers from outliers
- Runtime is non-deterministic and depends on match quality

The Stitching Pipeline: Merging & the Structural Bottleneck

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Warping & Blending

- Linear alpha blending removes visible seams between frames
- Heavy, vectorizable array manipulation

Feature Re-extraction

- Must run again on the updated canvas after every merge
- Canvas area grows monotonically:
$$A_n \approx A_0 + \sum_{i=1}^n \Delta A_i$$
- **Strict dependency:** stitch i cannot start until stitch $i - 1$ is committed

This rigid, monotonically-growing sequential chain is the central obstacle every parallel architecture in this work has to work around

Targeted Parallel Pipeline



Addresses each bottleneck according to its computational nature.

- **Threaded I/O:** image loading uses a `ThreadPoolExecutor`, since file reads release the GIL
- **Process-pool SIFT:** CPU-bound extraction uses independent processes to fully bypass the GIL
 - Requires a pickling workaround. C++ `cv2.KeyPoint` objects are decomposed into plain tuples before crossing the process boundary
- **Tiled Blending:** the output canvas is split into horizontal tiles blended concurrently
 - NumPy array arithmetic releases the GIL, so threads are highly efficient here too

Producer-Consumer Paradigm



Decouples the independent extraction work from the sequential fold:

- **Temporal overlap:** SIFT extraction for image $i + 1$ runs concurrently with the fold of image i
- **Architecture:** a producer thread dispatches every image to a warm process pool and pushes results onto a bounded queue. Queue, strictly in order
- **Zero-copy handoff:** producer and consumer share the same process address space, so passing an item costs only an object reference
- **Backpressure:** a fixed queue depth blocks the producer if it runs ahead of the consumer. This caps memory usage

MapReduce Paradigm



Restructures the fold into a binary merge tree:

- **Map phase:** SIFT extraction for all n images runs concurrently
- **Reduce phase:** adjacent pairs such as $(0, 1)$ and $(2, 3)$ are stitched concurrently. The process repeats over $\lceil \log_2 n \rceil$ levels instead of $n - 1$ sequential steps
- **The trade-off:** pairs are fixed by index rather than verified spatial overlap. This yields the highest peak speedup, but the result becomes geometrically unstable when a fixed pairing does not actually overlap

IPC Optimization Strategies



Pickling roughly 13.5 MB image arrays across process boundaries is expensive

- **Joblib:** the `loky` backend keeps worker processes alive across calls and automatically memory-maps arrays larger than 1 MB to disk instead of pickling them
- **Shared Memory:** images are copied once into a `multiprocessing.shared_memory` block. Workers then reconstruct a NumPy view instead of receiving a pickled copy
 - Only the outbound transfer is optimized. Results still return via the normal pickled channel

Performance Results

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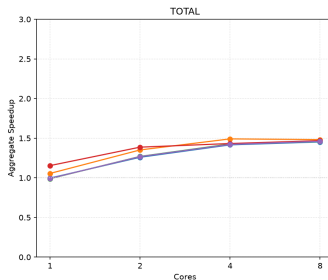
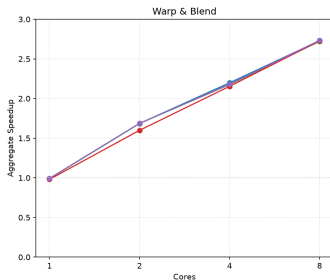
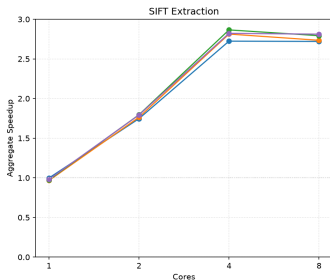
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Results: The Amdahl Ceiling

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Aggregate speedup vs. core count

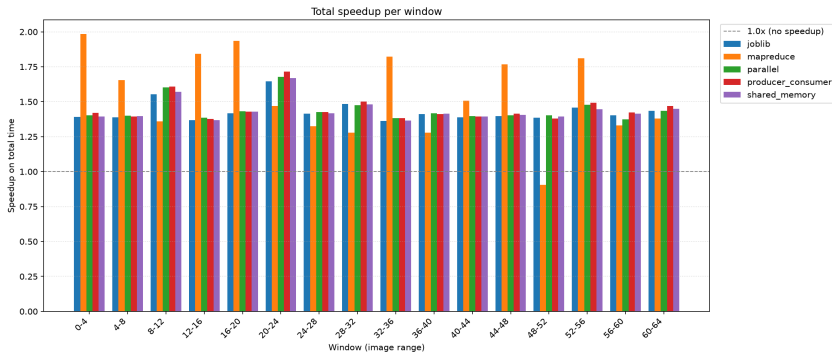
—●— joblib —●— mapreduce —●— parallel —●— producer_consumer —●— shared_memory



- **SIFT Extraction** plateaus at roughly $2.7\text{--}2.9\times$ past 4 cores, due to task starvation with a small window
- **Warp & Blend** scales almost linearly to 8 cores. This memory-bound work benefits from hyper-threading
- **TOTAL** never exceeds roughly $1.5\times$. Feature re-extraction takes about 65% of total time and stays fixed at approximately 14.5 s: it is never parallelized by any architecture

Results: Stability & the MapReduce Trade-off

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- **Linear-fold pipelines** are highly stable at $1.3\times$ to $1.7\times$ on every window
- **MapReduce** peaks near $2.0\times$ but drops to $0.91\times$
- **Why:** tree pairing is fixed by index rather than verified spatial overlap, so a geometrically unlucky pair has no fallback

Key Numbers

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Isolated Phases

- SIFT Extraction peaks at **2.9×** at 4 cores, then plateaus
- Warp & Blend is still rising at 8 cores, reaching **2.7×**

End-to-End

- TOTAL speedup is capped at **roughly 1.5×**
- MapReduce has the lowest average time, 21.1 s, and the highest peak, **2.0×**, but is the least predictable
- The four IPC-optimized linear-fold variants differ by **under 1%** from each other

Conclusions



- Isolated CPU-bound and I/O-bound tasks overcome the GIL cleanly with targeted multiprocessing and threading
- **IPC optimization gives diminishing returns** at this scale. Engineering effort on serialization, via Joblib and shared memory, yielded under 1% gain over the simplest process-pool baseline
- **Main Lesson:** true end-to-end scalability needs functional decomposition, as in MapReduce's merge tree, rather than retrofitting concurrency onto a rigid sequence. This comes at the cost of predictability
- **Future Work:** algorithmic restructuring of the re-extraction phase or offloading the sequential chain to GPUs

Parallel Image Stitching

Thank you for listening!
Any questions?

Per-Phase Speedup as Cores Increase

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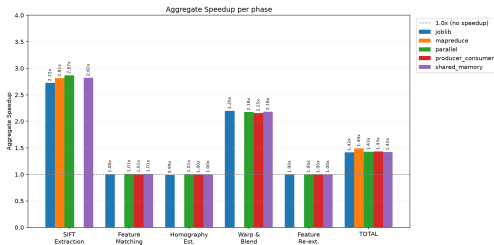


Figure: 4 cores

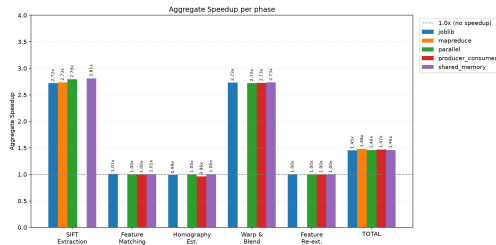


Figure: 8 cores

- Match, Homography and Re-extraction stay flat at about $1.0\times$ at every core count. None of the six pipelines ever parallelizes them
- At 8 cores with window size 4, only 4 tasks feed an 8-worker pool. This looks like a hardware plateau, but is it?

Task Starvation, Not a Hardware Limit

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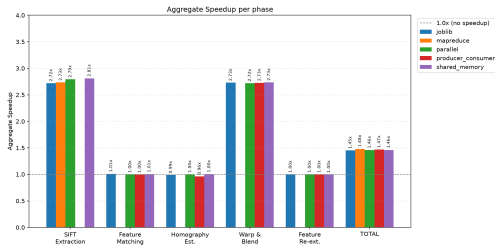


Figure: 8 cores, window size 4

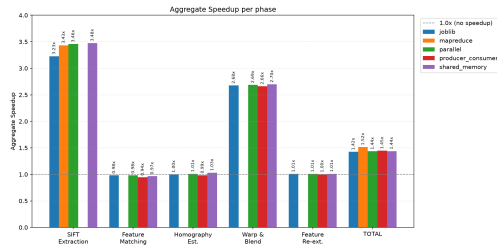


Figure: 8 cores, window size 8

- With window size 4, only 4 tasks feed an 8-worker pool. Half the pool sits idle, and dispatch overhead is disproportionate to the actual computation
- Doubling the window to 8 images fully populates the pool. SIFT Extraction breaks past the 2.7–2.8× plateau and reaches roughly 3.2–3.5×
- **Confirms** that the earlier plateau was task starvation, not a hardware ceiling. Hyper-threading limits still apply, but only once the pool is actually full